provisioned with *minikube*. Later in the book you'll use a managed Kubernetes service offered by a cloud provider.

In our daily job as developers, we don't want to spend too much time on infrastructural concerns, but it's critical to know the basics. Kubernetes has become the de facto orchestration tool and the common language for talking about containerized deployments. Cloud vendors have been building platforms on top of Kubernetes to provide a better experience for developers. Once you know how Kubernetes works, it'll be straightforward to use those platforms, because you'll be familiar with the language and the abstractions.

This chapter will walk you through the main features of Kubernetes and teach you how to create and manage Pods, Deployments, and Services for your Spring Boot applications. Along the way you'll enable graceful shutdown for your applications, learn how to scale them, and learn how to use the service discovery and load balancing features provided by Kubernetes. You'll also learn to automate your local development workflow with Tilt, visualize your workloads with Octant, and validate your Kubernetes manifests.

NOTE The source code for the examples in this chapter is available in the Chapter07/07-begin and Chapter07/07-end folders, which contain the initial and final states of the project (https://github.com/ThomasVitale/cloud-native-spring-in-action).

7.1 Moving from Docker to Kubernetes

With Docker Compose, you can manage the deployment of several containers at once, including the configuration of networks and storage. That is extremely powerful, but it's limited to one machine.

Using Docker CLI and Docker Compose, the interaction happens with a single Docker daemon that manages Docker resources on a single machine,